

User Guide



Dry Root Rot Portal (DRoP) - emphasis on chickpea



► **The Threat:** Dry Root Rot (DRR) is a highly destructive, emerging root disease caused by the hemibiotrophic fungus *Macrophomina phaseolina*. It primarily affects legumes (e.g., chickpea, soybean, mung bean) in drought-prone, low-rainfall regions, causing root necrosis, severe root loss, and plant death, which manifests as patches of dead plants and leads to massive yield losses.

► **The Challenge:** The pathogen has an exceptionally broad host range, infecting over 500 plant species. Its spread and host susceptibility are heavily driven by drought and adverse environmental/edaphic factors, posing an escalating threat to economically vital crops like chickpea.

► **The Solution:** Addressing this disease requires an integrated approach using innovative, intelligence-driven interventions. Currently, critical data on DRR—such as pathogen biology, epidemiology, host resistance, and management strategies—is scattered or unavailable. Therefore, there is an urgent need to build a consolidated, accessible repository of this information for the scientific community.

- Dr. Muthappa Senthil-Kumar
- Ms. Durgadevi Athimoolam
- Dr. Shikha Rani
- Dr. Rubi Jain
- Mr. Navaneeth Vinayachandran
- Mr. Shankar Acharya

Home page

External secondary links and biological databases curated to assist your research in Dry Root Rot disease.



Dry Root Rot Portal (DRoP) - emphasis on chickpea



Home About Tools Resources OMICS Help Contact

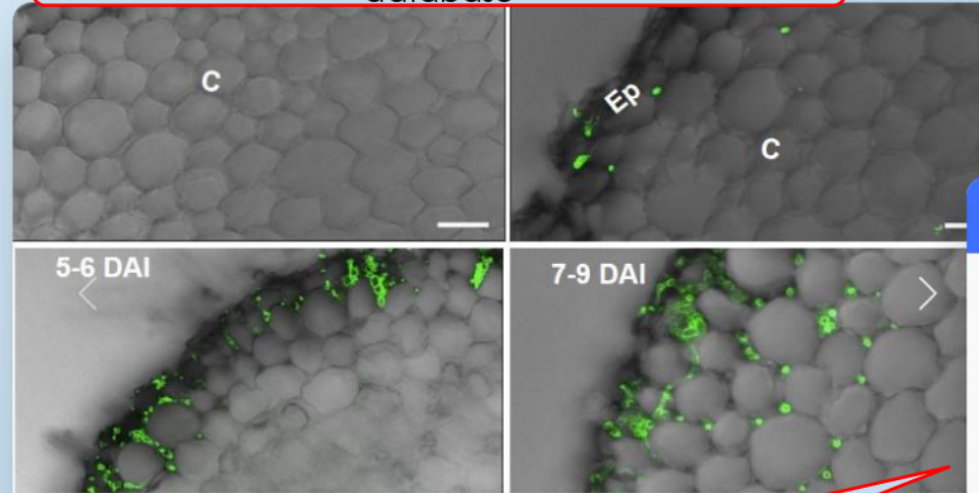
Navigation panel to seamlessly browse the database

Dry Root Rot Portal

Dry root rot (DRR) is an emerging and highly destructive root disease affecting a wide range of crops, primarily legumes like chickpea, soybean, mung bean, pigeon pea, groundnut etc., especially in drought-prone regions and low-rainfall seasons. This disease is caused by hemibiotrophic fungus *Macrophomina phaseolina*, which eventually leads to straw-yellow leaves, root necrosis, severe root loss, and ultimately plant death ([Irulappan et al., 2022](#); [Sinha et al. 2019](#); [Sinha et al. 2021](#)). In farm fields, the disease typically occurs as small patches of dead plants, resulting in huge yield losses ([Mirchandani et al. 2023](#)).

Currently, critical information related to this disease-including its host range, pathogen biology, disease epidemiology, host germplasm availability, environmental drivers, management strategies, mechanisms of host resistance, pathogen virulence, and advanced research strategies is scattered across disparate sources, while a significant amount of information remains unavailable. Hence, there is a need for a consolidated repository where all these resources and information can be systematically assembled and made accessible to the scientific community....

[Read More →](#)



Confocal-imaging of a transverse section of the chickpea root shows the colonization by *Macrophomina*. Ep-epidermis, C-cortex, DAI-days after infection.

Acts as a digital assistant to instantly interpret your queries, process information, and provide a helpful response.

Quick links

Github resources

Related links

DROP Assistant

Hello 🤖 I am your DROP assistant.
How can I help you today?

[What is DRR?](#) [Symptoms](#) [Management](#)

⚠️ AI-generated responses may contain errors. Please verify information.

Type your question here...

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DROP

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Dry Root Rot Portal

Dry root rot (DRR) is an emerging and highly destructive root disease affecting a wide range of crops, primarily legumes like chickpea, soybean, mung bean, pigeon pea, groundnut etc., especially in drought-prone regions and low-rainfall seasons. This disease is caused by hemibiotrophic fungus *Macrophomina phaseolina*, which eventually leads to straw-yellow leaves, root necrosis, severe root loss, and ultimately plant death (Irulappan et al., 2022; Sinha et al., 2019; Sinha et al., 2021). In farm fields, the disease typically occurs as small patches of dead plants, resulting in huge yield losses (Mirchandani et al., 2023).

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Brief introduction about the database and its objectives.

Visualization of global geographical distribution and epidemiological variations of DRR stress data



Chickpea genotype

- Open-source development repositories hosting machine learning models and curated databases.
- External secondary links and biological databases curated to assist your research in Dry Root Rot disease.
- The impact of this pathogen on major non-legume crops
- Foster national and international research collaborations.
- Hosts important scientific literature, peer-reviewed publications, and their hyperlinks pertaining to DRR stress in plants.

- Quick links
- Github resources
- Related links
- DRR in other crops
- Data statistics
- FAIR Principles
- Author repository
- Lab publications
- Feedback

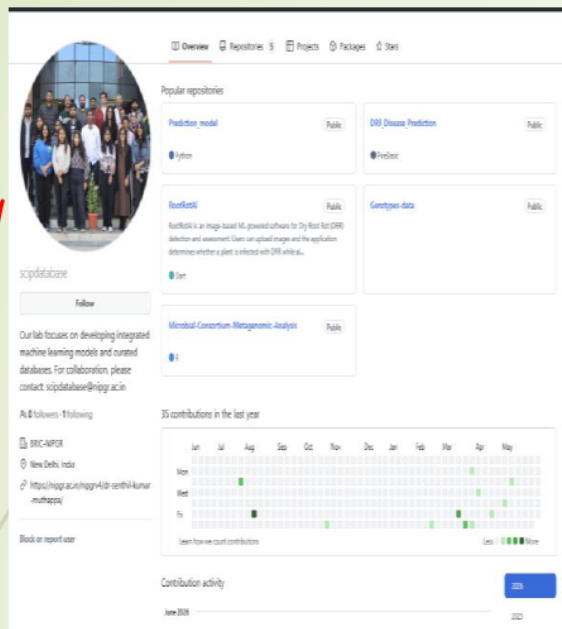
Geospatial/Stress map

Sitemap

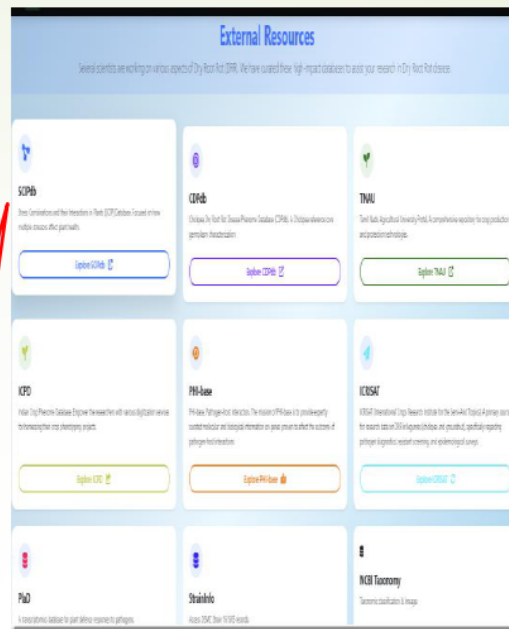
Directory path access to Dry Root Rot database tools and resources

Users are encouraged to share feedback about their experience with the database to help us improve.

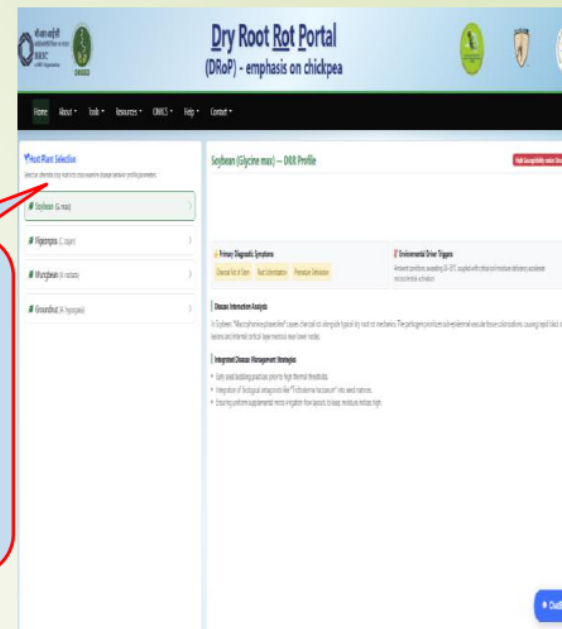
Open-source development repositories hosting machine learning models and curated databases.



Foster national and international research collaborations



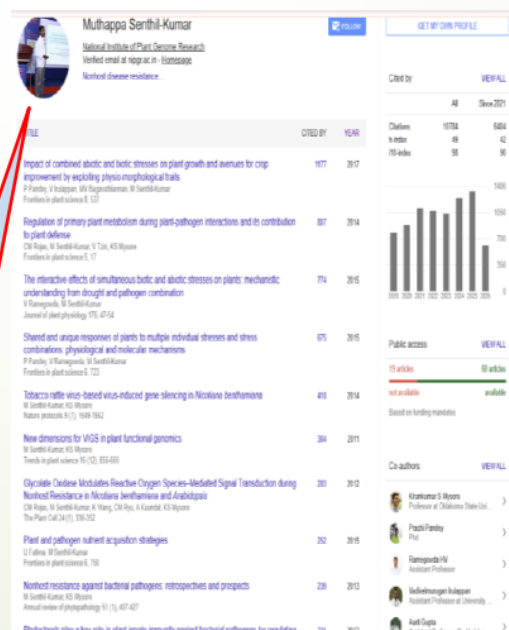
The impact of this pathogen on major non-legume crops



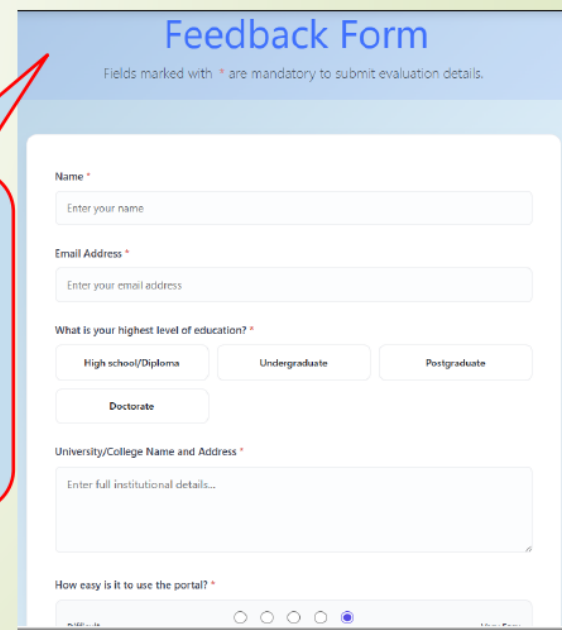
Foster national and international research collaborations



Hosts important scientific literature, peer-reviewed publications, and their hyperlinks pertaining to DRR stress in plants.



Users are encouraged to share feedback about their experience with the database to help us improve.





Geographical distribution description

Colored based on the net outcome

- 1) A green icon indicates a major research institute worldwide where studies on dry root rot disease are performed
- 2) A blue icon indicates a major research institute in India where rainfall and disease studies are conducted.
- 3) A brown icon indicates a major research institute in India where year-by-year disease studies have been conducted over the last 10 years.

DRoP Portal Navigation Map	
Directory path access to Dry Root Rot database tools and resources	
NAVIGATION LINK	DESCRIPTION
HOME	
What is Dry Root Rot Portal	Brief introduction about the database and its objectives.
Slider Banner	Representative images illustrating the effect of DRR stress on plants.
Github Resources	Open-source development repositories hosting machine learning models and curated databases.
Other Related Links	External secondary links and biological databases curated to assist your research in Dry Root Rot disease.
Author Repository	Foster national and international research collaborations.
Publication	Hosts important scientific literature, peer-reviewed publications, and their hyperlinks pertaining to DRR stress in plants.
Copyright	Licensing frameworks, policies, and legal parameters for data reuse and sharing.
Geospatial/Stress Map	Visualization of global geographical distribution and epidemiological variations of DRR stress data.
DRoP Assistant	Acts as a digital assistant to instantly interpret your queries, process information, and provide a helpful response.
ABOUT	
Disease Description	Comprehensive symptoms, etiology, and global epidemiological effects of Dry Root Rot.
Pathogen Description	Morphological characteristics, lifecycle, and virulence factors of <i>Macrophomina phaseolina</i>
Management Systems	Information about strategic crop security and control measures.

Directory path access to Dry Root Rot database tools and resources

About



Dry Root Rot Portal (DRoP)

kpea focused resource



Comprehensive symptoms, etiology, and global epidemiological effects of Dry Root Rot.

Morphological characteristics, lifecycle, and virulence factors of *Macrophomina phaseolina*

External secondary links and biological databases curated to assist your research in Dry Root Rot disease.

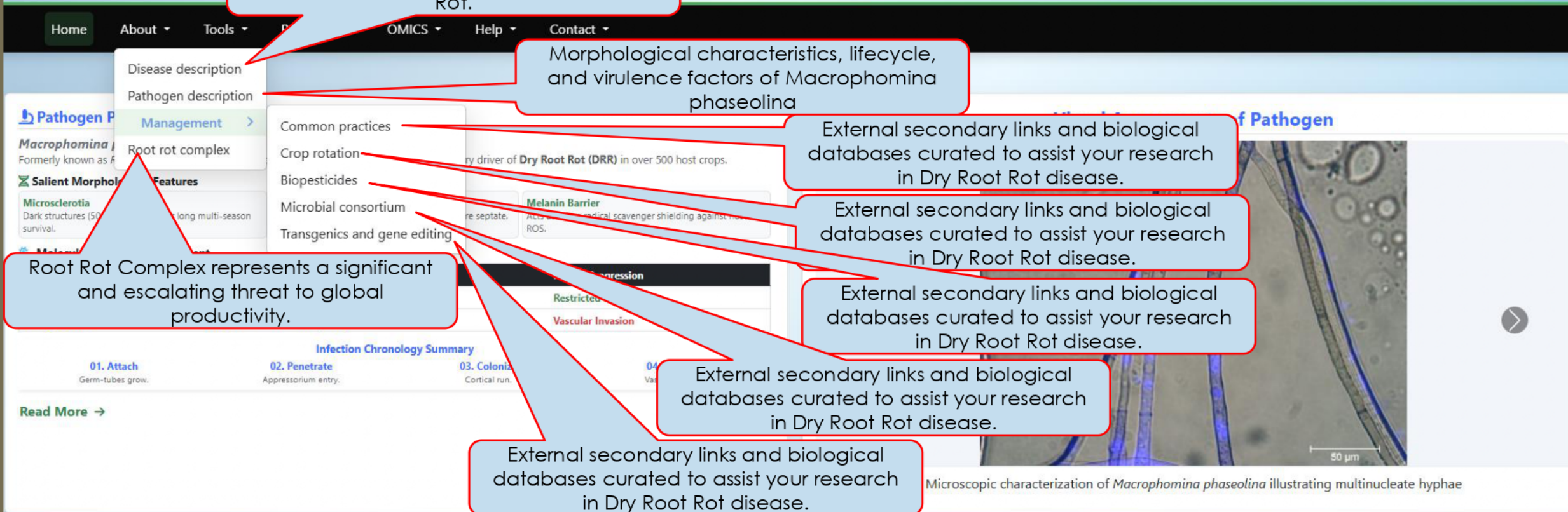
External secondary links and biological databases curated to assist your research in Dry Root Rot disease.

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External secondary links and biological databases curated to assist your research in Dry Root Rot disease.

Root Rot Complex represents a significant and escalating threat to global productivity.



DRoP

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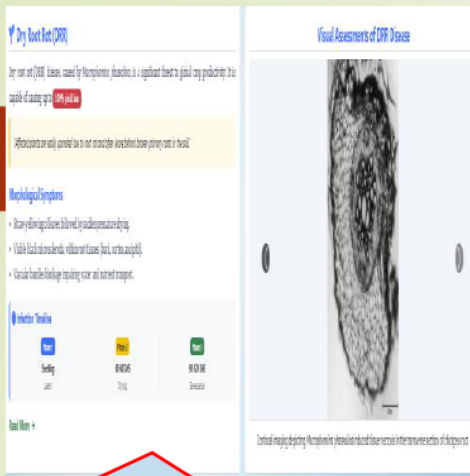
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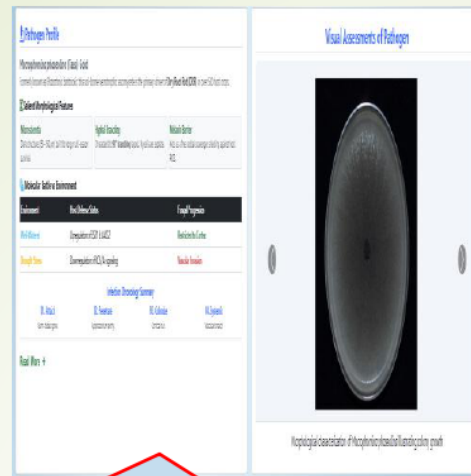
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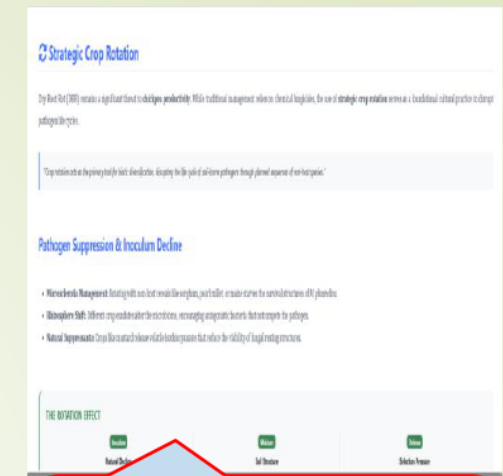
Comprehensive symptoms, etiology, and global epidemiological effects of Dry Root Rot.



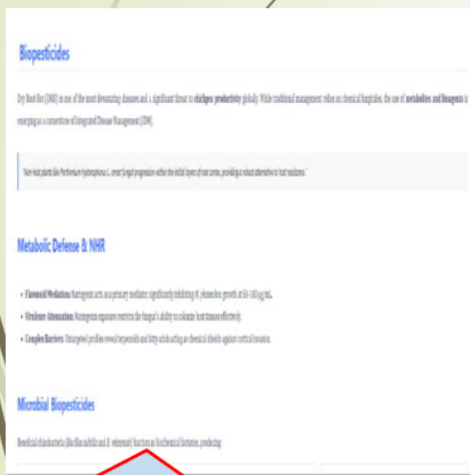
Morphological characteristics, lifecycle, and virulence factors of *Macrophomina phaseolina*



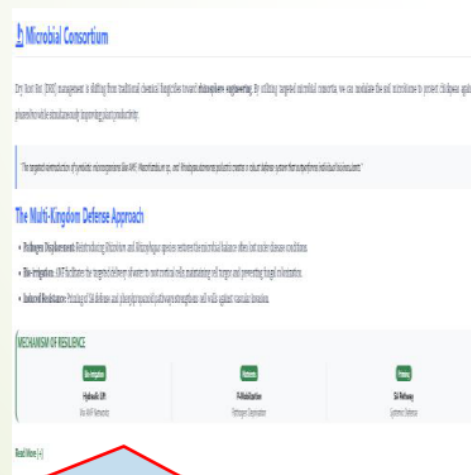
Traditional control mechanisms and field routines.



Anti-fungal proteins offer strong fungicidal properties by disrupting fungal cell membranes and reducing metabolic acidification.



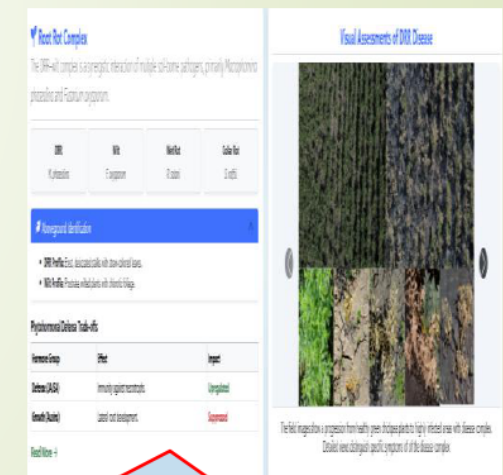
The use of bio pesticides emerging as a cornerstone of Integrated Disease Management (IDM).



Crop rotation acts as the primary tool for biotic diversification, disrupting the life cycle of soil-borne pathogens through planned sequences of non-host species.



Microbial consortium creates a robust defense system that outperforms individual bioinoculants.



Root Rot Complex represents a significant and escalating threat to global productivity.

Tools

Dry Root Rot Portal (DRoP) - emphasis on chickpea

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Disease identification and assessment

Numerical data based

Image data based

RootRotAI-2.0 (Github repo)

RootRotAI-2.1 (Web_app)

RootRotAI-2.2 (Mobile_app)

Integrated deep learning validation tools and diagnostic pipelines.

Deep-learning vision diagnostics built for multi-stage symptom evaluation from image scans.

Supports three modalities: camera, root scanner, and microscope to determine whether a plant is infected with DRR while also assigning a disease severity score ranging from 1 to 5, where 5 is highly susceptible

Artificial neural network framework tracking environmental factors to assess probability of the disease.

As a multitask Vision Transformer designed for simultaneous binary classification (DRR vs. Control) and regression-based severity estimation

Utilizes two multitask transformer-based TensorFlow Lite models, one optimized for camera and root scanner modalities, and another for microscope images.

Quick links

- Github resources
- Related links
- DRR in other crops
- Data statistics
- FAIR Principles
- Author repository
- Lab publications
- Feedback

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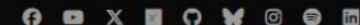
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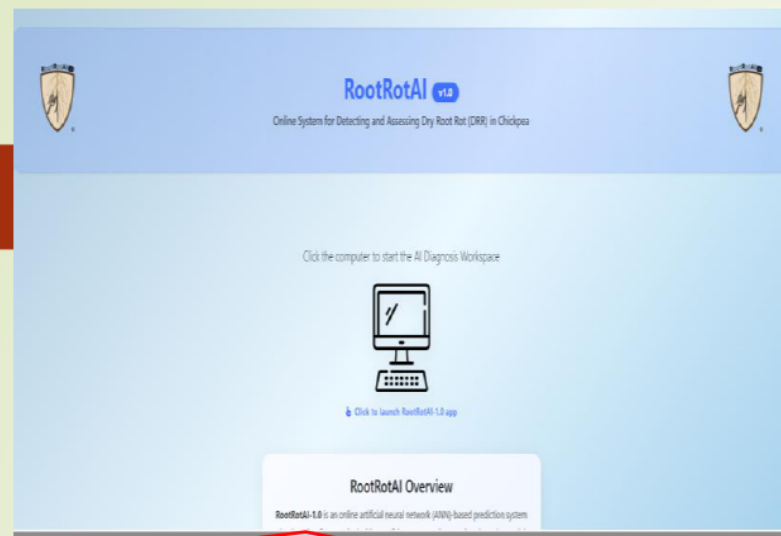
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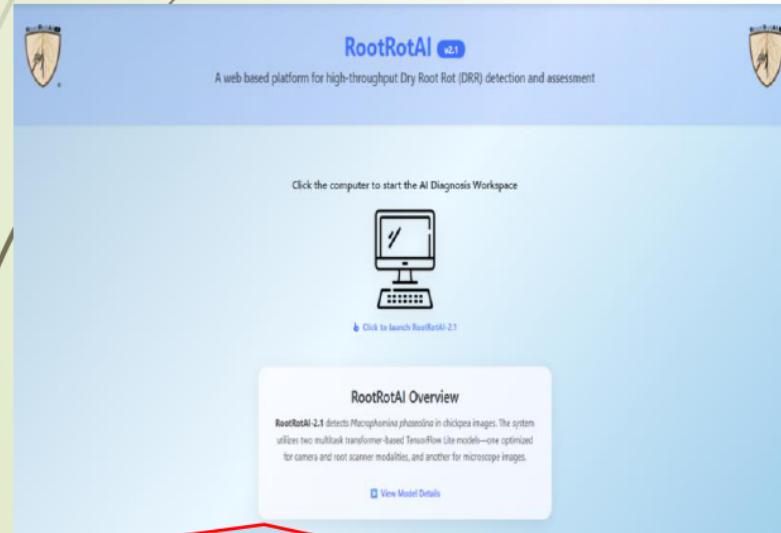


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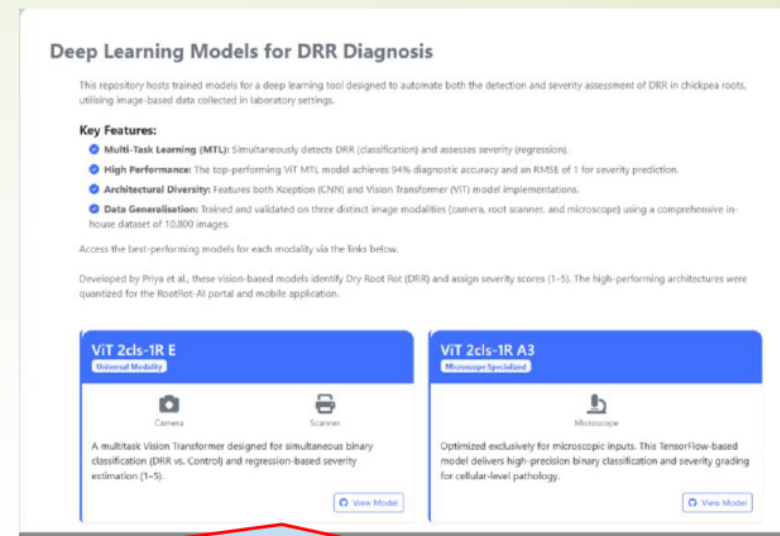
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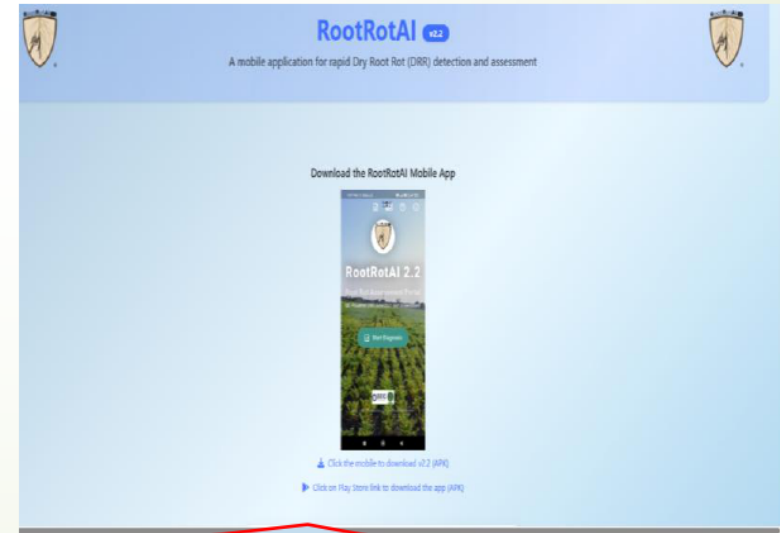
Artificial neural network framework tracking environmental factors to assess probability of the disease.



Utilizes two multitask transformer-based TensorFlow Lite models, one optimized for camera and root scanner modalities, and another for microscope images.



A multitask Vision Transformer designed for simultaneous binary classification (DRR vs. Control) and regression-based severity estimation (1–5).



Supports three modalities: camera, root scanner, and microscope to determine whether a plant is infected with DRR while also assigning a disease severity score ranging from 1 to 5, where 5 is highly susceptible

Resources



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Dry Root Rot Portal

Dry root rot (DRR) is an emerging and highly destructive disease of chickpea, primarily legumes like chickpea, soybean, and mungbean, in drought-prone regions and low-rainfall areas. It is caused by the soil-borne fungus *Macrophomina phaseolina*, which eventually leads to plant death (Irulappan et al., 2023).

Currently, critical information related to this disease, including its host range, pathogen diversity, epidemiology, host genetic resistance, environmental factors, management strategies, and advanced breeding approaches, is scattered across disparate sources. Hence, there is a need for a centralized platform where such information can be assembled and made accessible to researchers and breeders.

Protocol

Pathogen strains

Literature

Germplasm

Images

Videos

DRR screening facility

Standard Operating Procedures (SOPs) for isolation, cultural growth, and seedling validation.

Detailed information about pathogen strains pertaining to DRR stress in plants.

Curated referencing and links pertaining to DRR stress in plants.

Detailed information about accessions related to chickpea.

Classifications assigned based on the severity of the DRR disease observed.

High-resolution image visualizing Dry Root Rot (DRR) progression through multiple imaging techniques.

A curated list of videos from the DROP database focusing on the impact of Dry Root Rot disease.

An integrated platform for rapid generation advancement and trait-based phenotyping for chickpea improvement.

Passport collection

Mini-core collection

Pathogen strains

Images

Videos

DRR screening facility

Geospatial/Stress map

Sitemap

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Dry Root Rot Portal

Represents a database presents the phenotypic characteristics and comprises several traits studied under DRR.

including its host pathogen biology, disease epidemiology, host germplasm availability, environmental drivers, mechanisms of host resistance, pathogen virulence, and advanced research strategies, while a significant amount of information remains unavailable. A consolidated repository where all these resources and information can be made accessible to the scientific community....

Read More

Provide mechanistic insight into hormone-mediated resistance responses and secondary pathways in chickpea.

Genomics > Plant genomics

Metagenomics
Transcriptomics
Proteomics
Metabolomics
Phenomics

Pathogen genomics

A range of crops, etc., especially in biotrophic fungus, root loss, In farm

pathogen biology, disease strategies, mechanisms across dispa a need f

Studies performed using a chickpea reference core germplasm collection

Accessing sequence-linked metadata and facilitates downstream analyses in epidemiology, population genomics, and host-pathogen interaction studies.

Profiling of chickpea rhizosphere to reveal complex shifts in microbial community dynamics under dry root rot and drought stress

Functional genomics studies aimed at identifying key genes, pathways, and regulatory networks involved in dry root rot pathogenesis and resistance

Quantitative analysis mapping protein level accumulation profiles across stress boundaries.

Sick pot experiment under greenhouse conditions

Geospatial/Stress map

Sitemap

GitHub resources

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Plant Genomics

Genome-wide association studies (GWAS) were performed using a chickpea reference core germplasm collection consisting of 355 genotypes, including 254 landraces, 38 improved varieties, and 18 breeding lines, representing diverse geographical regions across Asia, Africa, Europe, North America, and South America. The collection included 221 dea, 74 kaali, and 11 intermediate genotypes. Through GWAS, identified 12 million high-quality SNPs and 11 significant marker-trait associations (MTAs) for disease severity index under dry root rot, osmotic, and combined stress (DPR + osmotic stress) responses. A total of 13 MTAs were identified under DRR stress, while 14 stress-responsive candidate genes were identified under combined stress. The associated loci included genes related to disease resistance, transcription factors, receptor-like kinases (RLKs), protein transporters, and sugar transporters. Notably, *GaMEST18*, a sugar transporter gene, showed a strong association with susceptibility to DRR and combined stress. Functional validation demonstrated that overexpression of *GaMEST18* increased susceptibility to *Masovirina* phoxoselin, whereas RNAi-mediated silencing enhanced tolerance. Overall, this study provides important insights into the genetic architecture of DRR susceptibility under osmotic stress and highlight *GaMEST18* as a promising target for breeding DRR-resistant chickpea cultivars (Rangan et al., 2025, under revision).

Studies performed using a chickpea reference core germplasm collection

Pathogen Genomics

The genomic landscape of *Masovirina* sp. reveals substantial strain-level diversity across host species and geographic regions, highlighting the need for robust diagnostic markers and effective disease management strategies. For example, Michalewicz and Kumar (unpublished) reported that the taxonomic identity of the dry root rot pathogen has historically been confused with *Rhizoctonia solani* (Tulas), likely due to reliance on morphology-based classification, and the rare observation of the sporulating stage in plants. Genomic and phylogenomic analyses have now confirmed that the DRR pathogen belongs to the family *Botryosphaeraceae* (Ascomycota) and is a distinct from *Rhizoctonia* like basidiomycetes. A large-scale population genomics study by [Saxena et al. \(2022\)](#) analyzed draft-level sequencing data from 463 *Masovirina* isolates sampled from 58 host plant species and soils across 20 countries, revealing high genomic diversity, extensive admixture, and near-equal mating-type ratios, consistent with ongoing recombination.

Kingdom: Fungi, Phylum: Ascomycota, Family: Botryosphaeraceae, Genus: *Masovirina*, Species: *M. phoxoselin*

Q Use the Search box to filter genome data by keyword. Click on the column headers to sort results.

Show 10 entries

S.No.	Assembly Name	Assembly Location	Pathogen Name	Strain Name	Subproject	Strain Sample	Host Name	Location	RNA Accession	Reference
1	NA	NA	<i>Masovirina phoxoselin</i>	CC2465	PDRAC2465	PDRAC2465	Chickpea (Cicer arietinum)	Aligarh, India	PDRAC2465	PDRAC2465
2	NA	NA	<i>Masovirina phoxoselin</i>	CC2466	PDRAC2466	PDRAC2466	Chickpea (Cicer arietinum)	Chitkoti, India	PDRAC2466	PDRAC2466
3	NA	NA	<i>Masovirina phoxoselin</i>	CC2467	PDRAC2467	PDRAC2467	Chickpea (Cicer arietinum)	Chennai, India	PDRAC2467	PDRAC2467
4	NA	NA	<i>Masovirina phoxoselin</i>	CC2468	PDRAC2468	PDRAC2468	Chickpea (Cicer arietinum)	Andhra, India	PDRAC2468	PDRAC2468
5	NA	NA	<i>Masovirina phoxoselin</i>	CC2469	PDRAC2469	PDRAC2469	Chickpea (Cicer arietinum)	Madhya, India	PDRAC2469	PDRAC2469

Accessing sequence-linked metadata and facilitates downstream analyses in epidemiology, population genomics, and host-pathogen interaction studies.

Metagenomics

Metagenomic profiling of the chickpea rhizosphere was performed to reveal complex shifts in microbial community dynamics under dry root rot and drought stress. Using both amplicon (16S rRNA and ITS) and whole metagenome sequencing, the rhizosphere microbiome was found to be predominantly bacterial (~90%), with minor contributions from archaea, fungi, algae, and viruses. Under combined drought and disease stress, a significant shift was observed, wherein fungal and archaeal communities increased, while bacterial, algal, and viral communities were suppressed. Pathogenic fungal species were notably enriched under drought conditions, whereas symbionts were favored and saprotrophs were suppressed. Among the major fungal root pathogens, *Masovirina phoxoselin*, *Fusarium solani*, and *Rhizoctonia solani* increased abundance, whereas *Ascomycota* and *Basidiomycota* communities exhibited reduced abundance in soil under drought stress.

Amplicon sequencing (16S rRNA and ITS) identified 46 bacterial and 38 fungal operational taxonomic units (OTUs) and revealed a high abundance of *M. phoxoselin* pathogen stress, which was further exacerbated by drought. Drought was found to adversely affect root colonization by beneficial saprotrophs and symbionts while promoting an increase in pathogenic fungi. Furthermore, the relative abundance of *Ascomycota* and *Basidiomycota* increased under drought stress, whereas colonization of viral endosymbionts, nitrogen-fixing bacteria was significantly reduced. Finally, when combined drought and disease stress were compared with disease complex only treatments, fungal diversity as reflected by OTU richness, increased, while bacterial diversity declined. These findings highlight how the convergence of drought and fungal pathogens alters the rhizosphere microbial landscape, favoring disease progression and suppressing beneficial symbiotic associations ([Chakraborty et al. 2023](#)).

Q Use the Search box to filter metagenome data by keyword. Click on the column headers to sort results.

Show 10 entries

S.No.	Experiment ID	Host Genotype	Accession	Location	Stress treatment	Library	Run Accession	Sample type	RNA file
1	PDRAC2465	Chickpea (Cicer arietinum)	PDRAC2465	PDRAC2465	Combined stress (Disease complex + Drought)	Next	PDRAC2465	Rhizosphere metagenome	PDRAC2465
2	PDRAC2466	Chickpea (Cicer arietinum)	PDRAC2466	PDRAC2466	Combined stress (Disease complex + Drought)	Next	PDRAC2466	Rhizosphere metagenome	PDRAC2466
3	PDRAC2467	Chickpea (Cicer arietinum)	PDRAC2467	PDRAC2467	Combined stress (Disease complex + Drought)	Next	PDRAC2467	Rhizosphere metagenome	PDRAC2467
4	PDRAC2468	Chickpea (Cicer arietinum)	PDRAC2468	PDRAC2468	Combined stress (Disease complex + Drought)	Next	PDRAC2468	Rhizosphere metagenome	PDRAC2468
5	PDRAC2469	Chickpea (Cicer arietinum)	PDRAC2469	PDRAC2469	Combined stress (Disease complex + Drought)	Next	PDRAC2469	Rhizosphere metagenome	PDRAC2469

Profiling of chickpea rhizosphere to reveal complex shifts in microbial community dynamics under dry root rot and drought stress

Transcriptomics

Transcriptomic profiling of chickpea roots under drought, pathogen, and combined drought-pathogen stress at 3 and 5 days after sowing (DAS) revealed extensive transcriptional reprogramming, with a higher number of differentially expressed genes at 1 DAS, likely associated with the transition of dry root rot (DRR) from a biotrophic to necrotrophic phase ([Saxena et al. 2023](#)). Further transcriptome analysis at 1 hour and 24 hours following water stress exposure captured early signaling genes involved in stress perception and water stress sensing, as well as downstream response, including genes associated with pathogenesis, root susceptibility, defense, and hormone signaling ([Sax et al. 2023](#)).

Dual RNA-seq of root samples from the biotrophic stage, collected at 3 and 5 days after inoculation (DAS) for *Perthapsium hyphomysum* and at 2 DAS for the biotrophic chickpea genotype CC-4558 showed that *P. hyphomysum* roots a strong immune response against *Masovirina phoxoselin*, indicative of post-invasive root resistance (NDR). Key responses included cell wall synthesis, phytohormone signaling, defense pathways, and phenylpropanoid biosynthesis, while the pathogen activated genes in the xenobiotic degradation pathway, highlighting the dynamic host-pathogen interaction. These findings suggest that general defense, hormone signaling, and secondary metabolism pathways collectively contribute to NDR in *P. hyphomysum* against DRR ([Chakraborty et al. 2023](#)).

Transcriptomic Profiling

Q Use the Search box to filter transcriptome data by keyword. Click on the column headers to sort results.

Show 10 entries

S.No.	Experiment ID	Host Name	Tissue	Genotype	Stage	Sample Type	Strain Name	Example Accession	TSA Accession	Reference
1	PDRAC2465	Perthapsium hyphomysum	Root	-	1 DAS	Pathogen	SWAPN	PDRAC2465	PDRAC2465	Michalewicz et al. 2023
2	PDRAC2466	Perthapsium hyphomysum	Root	-	2 DAS	Pathogen	SWAPN	PDRAC2466	PDRAC2466	Michalewicz et al. 2023
3	PDRAC2467	Perthapsium hyphomysum	Root	-	3 DAS	Pathogen	SWAPN	PDRAC2467	PDRAC2467	Michalewicz et al. 2023
4	PDRAC2468	Perthapsium hyphomysum	Root	-	4 DAS	Pathogen	SWAPN	PDRAC2468	PDRAC2468	Michalewicz et al. 2023
5	PDRAC2469	Perthapsium hyphomysum	Root	-	5 DAS	Pathogen	SWAPN	PDRAC2469	PDRAC2469	Michalewicz et al. 2023

Functional genomics studies aimed at identifying key genes, pathways, and regulatory networks involved in dry root rot pathogenesis and resistance

Proteomics

Proteomic profiling of the soybean root apoplast during infection by *M. phoxoselin* has provided direct molecular evidence of extracellular remodeling in response to the pathogen. By integrating apoplastic proteomics with AlphaFold structural annotations, this study characterized infection-associated changes within the host exosome. A total of 241 proteins carrying signal peptides were identified, including 98 CAZymes, 27 proteases, and 16 protease inhibitors. The remaining proteins represented diverse functions, including Antigen 5, C4P proteins, chaperone proteins, fasciculin-like arabinogalactans, pathogenesis-related 1 protein, chitinase, and other proteins.

Comparative analysis between control and infected samples indicated that 241 proteins were common to both conditions, whereas eight proteins were detected in control samples and six proteins were uniquely detected in infected samples. Quantitative analysis identified 25 apoplastic proteins that were differentially accumulated. Among these, 19 proteins showed increased accumulation and 19 proteins exhibited reduced abundance, while 146 proteins remained unchanged. Enrichment analysis revealed that proteins with laccase/oxidase activity, peptidase, ribonuclease, chitinase, and alpha-mannosidase activity, peptidase activity, and flavin adenine dinucleotide binding were predominantly enriched among proteins that increased in abundance during infection. Conversely, proteins involved in protein transport, protein transport, and protein transport were predominantly enriched among proteins that decreased in abundance during infection. Overall, these findings suggest extensive modulation of extracellular defense mechanisms and oxidative stress-related processes during dry root rot progression. Structural predictions provide a basis for prioritizing extracellular defense-associated candidates for functional validation and resistance engineering against *M. phoxoselin* in soybean ([2023](#)).

Host: Soybean, Tissue: Root apoplast, Total proteins identified: 255, Differential proteins: 95, Major protein classes: CAZymes, proteases, protease inhibitors, Protein 12, P10061044

Quantitative analysis mapping protein level accumulation on profiles across stress boundaries.

Metabolomics

Metabolomic profiling provides insight into the metabolic pathways involved in host-pathogen interactions, revealing metabolic remodeling during dry root rot progression. By integrating metabolomic data with AlphaFold structural annotations, this study characterized infection-associated changes within the host exosome. A total of 241 proteins carrying signal peptides were identified, including 98 CAZymes, 27 proteases, and 16 protease inhibitors. The remaining proteins represented diverse functions, including Antigen 5, C4P proteins, chaperone proteins, fasciculin-like arabinogalactans, pathogenesis-related 1 protein, chitinase, and other proteins.

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Q Use the Search box to filter metabolite data by keyword. Click on the column headers to sort results.

S.No.	Metabolite	Level Type	Metabolite ID	Accession	Location	Stress treatment	Library	Run Accession	Sample type	RNA file
1	Glucose	1 DAS	1000	1000	1000	1000	1000	1000	1000	1000
2	Glucose	1 DAS	1001	1001	1001	1001	1001	1001	1001	1001
3	Glucose	1 DAS	1002	1002	1002	1002	1002	1002	1002	1002
4	Glucose	1 DAS	1003	1003	1003	1003	1003	1003	1003	1003

Represents a database presents the phenotypic characteristics and comprises several traits studied under DRR.

Chickpea Dry Root Rot Disease Phenome Database (CDPdb)

Chickpea reference core germplasm characterization

CDPdb

Dry root rot (DRR) disease, an emerging threat to chickpea in the era of climate change, is caused by the soil-borne necrotrophic fungal pathogen *Masovirina phoxoselin* ([Chen et al. 2023](#), [Jain et al. 2023](#)). DRR disease incidence is aggravated by drought and high temperatures, causing severe yield losses of up to 10-100% in susceptible varieties ([Chakraborty et al. 2023](#), [Chakraborty et al. 2023](#), [Chakraborty et al. 2023](#)). Recognizing its significance, efforts have been made to identify sources of resistance in chickpea. Although several studies have been conducted, a complete source of resistance is currently unknown.

Genotype Statistics

Features	Total Number
Genotypes	305
Classes	3
Countries	25
Biological status	3
Total traits	8
Stress conditions	3

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Provide mechanistic insight into hormone-mediated resistance responses and secondary pathways in chickpea.

Help

Dry Root Rot Portal (DRoP) - emphasis on chickpea

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Step-by-step query navigation mapping of the content.

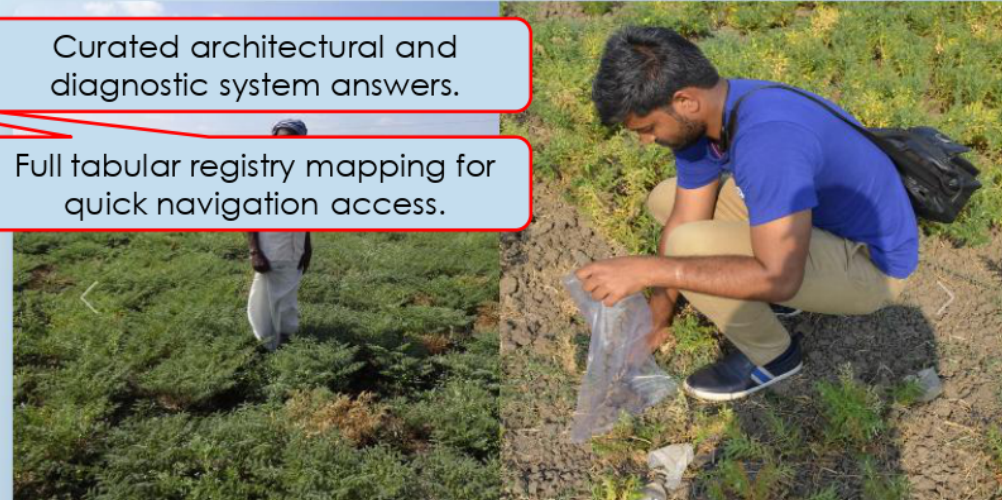
Curated architectural and diagnostic system answers.

Full tabular registry mapping for quick navigation access.

Dry Root Rot Portal

Dry root rot (DRR) is an emerging and highly destructive root disease affecting a wide range of crops, primarily legumes like chickpea, soybean, mung bean, pigeon pea, groundnut etc., especially in drought-prone regions and low-rainfall seasons. This disease is caused by hemibiotrophic fungus *Macrophomina phaseolina*, which eventually leads to straw-yellow leaves, root necrosis, severe root loss, and ultimately plant death ([Irulappan et al., 2022](#); [Sinha et al., 2019](#); [Sinha et al., 2021](#)). In farm fields, the disease typically occurs as small patches of dead plants, resulting in huge yield losses ([Mirchandani et al., 2023](#)).

Currently, critical information related to this disease-including its host range, pathogen biology, disease epidemiology, host germplasm availability, environmental drivers, management strategies, mechanisms of host resistance, pathogen virulence, and advanced research strategies is scattered across disparate sources, while a significant amount of information remains unavailable. Hence, there is a need for a consolidated repository where all these resources and information can be systematically assembled and made accessible to the scientific community....

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Left-View from farm field showing DRR plant disease, Right-Sample collection for scientific studies from DRR infected farmer field

 Geospatial/ Stress map

 Sitemap

Quick links

- [Github resources](#)
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- [Data statistics](#)
- [FAIR Principles](#)
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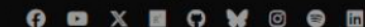
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User Guide



■ **The Threat:** Dry Root Rot (DRR) is a highly destructive, emerging root disease caused by the hemibiotrophic fungus *Macrophomina phaseolina*. It primarily affects legumes (e.g., chickpea, soybean, mung bean) in drought-prone, low-rainfall regions, causing root necrosis, severe root loss, and plant death, which manifests as patches of dead plants and leads to massive yield losses.

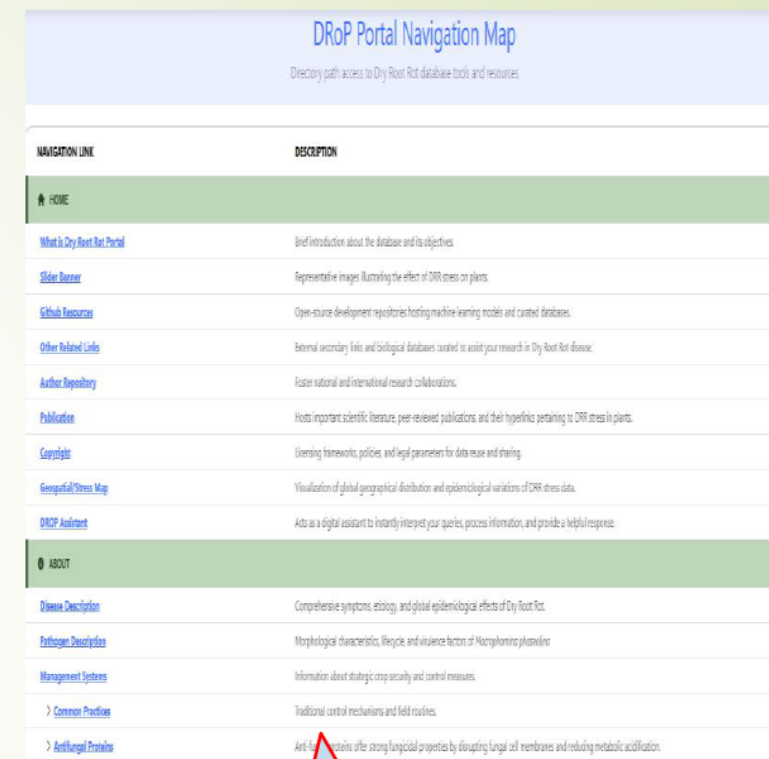
■ **The Challenge:** The pathogen has an exceptionally broad host range, infecting over 500 plant species. Its spread and host susceptibility are heavily driven by drought and adverse environmental/edaphic factors, posing an escalating threat to economically vital crops like chickpea.

■ **The Solution:** Addressing this disease requires an integrated approach using innovative, intelligence-driven interventions. Currently, critical data on DRR—such as pathogen biology, epidemiology, host resistance, and management strategies—is scattered or unavailable. Therefore, there is an urgent need to build a consolidated, accessible repository of this information for the scientific community.

Step-by-step query navigation mapping of the content.



Curated architectural and diagnostic system answers.



Full tabular registry mapping for quick navigation access.

Contact

Dry Root Rot Portal (DRoP) - emphasis on chickpea



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Dry Root Rot Portal

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Currently, critical information related to this disease-including its host range, pathogen biology, disease epidemiology, host germplasm availability, environmental drivers, management strategies, mechanisms of host resistance, pathogen virulence, and advanced research strategies is scattered across disparate sources, while a significant amount of information remains unavailable. Hence, there is a need for a consolidated repository where all these resources and information can be systematically assembled and made accessible to the scientific community....

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Team members

Funding & Acknowledgement

Reach us

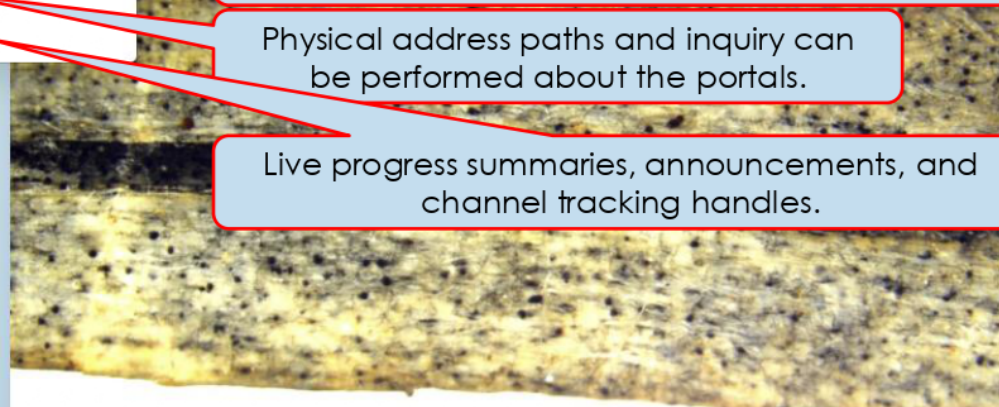
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Scientific minds behind the portal.

Financial support and acknowledgment for the scientists and researchers for their invaluable contributions during the development of this portal.

Physical address paths and inquiry can be performed about the portals.

Live progress summaries, announcements, and channel tracking handles.



Split open root image showing microsclerotia in the pith region of infected chickpea

Geospatial/Stress map

Sitemap

Github resources

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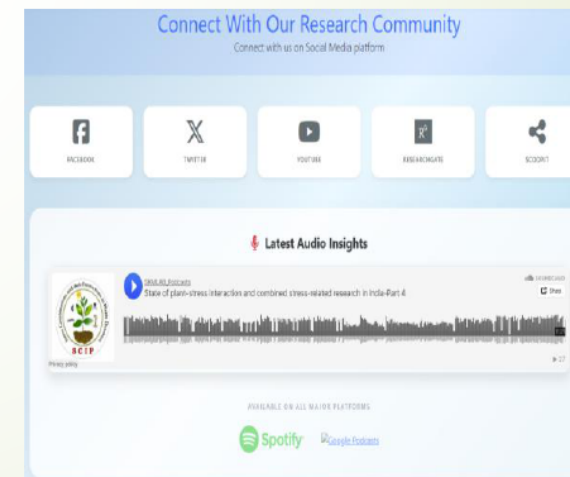
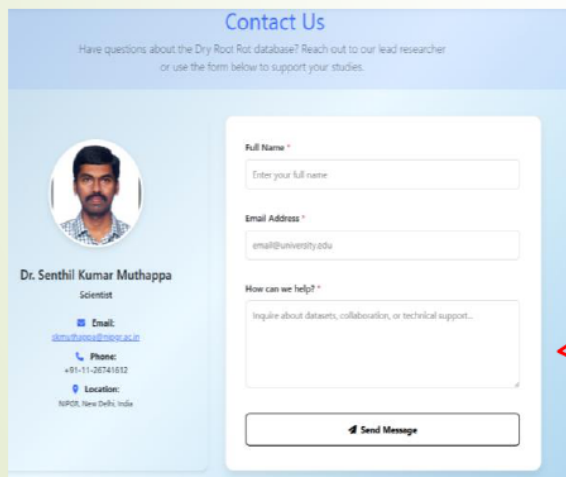
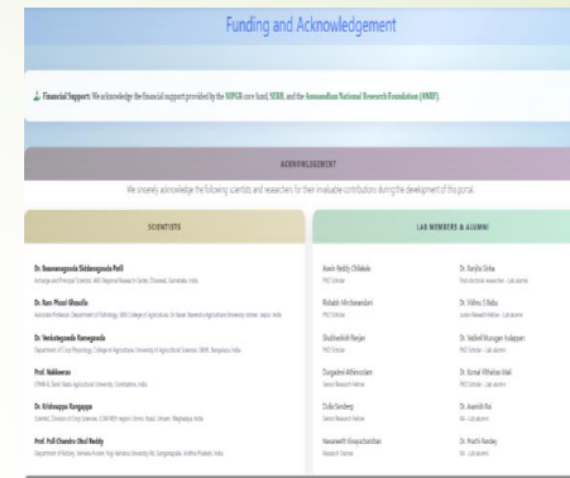
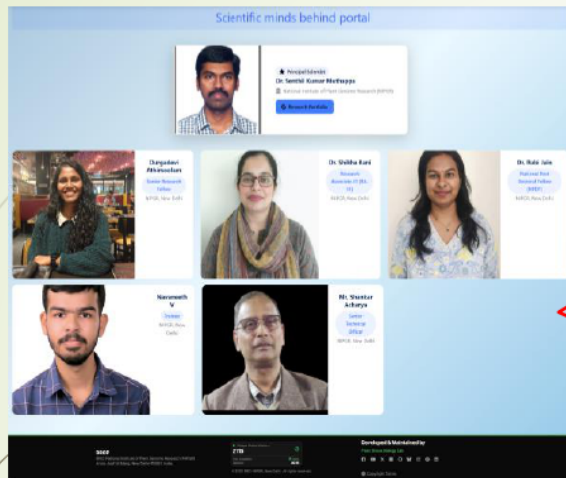
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Summarize the Portal

The document provides a comprehensive overview of Dry Root Rot disease, including its biology, epidemiology, management strategies, research tools, and collaborative efforts. Overview of Dry Root Rot Disease A detailed resource on Dry Root Rot caused by *Macrophomina phaseolina*, covering biology, spread, detection, and control. Dry Root Rot caused by *Macrophomina phaseolina* poses a significant threat to legumes and other crops, requiring integrated management and advanced diagnostics.

Disease Biology and Symptoms

- Describes the morphological features, lifecycle, and virulence factors of *Macrophomina phaseolina*.
- Highlights symptoms and progression of Dry Root Rot in affected plants.
- Emphasizes the root necrosis and plant death caused by the pathogen.

Epidemiology and Global Impact

- Discusses the pathogen's broad host range and environmental factors influencing spread.
- Presents data on the geographical distribution and variations of the disease.
- Notes the threat to major crops like chickpea, soybean, and mung bean.

Disease Management Strategies

- Recommends integrated approaches including crop rotation, bio-pesticides, and microbial consortia.
- Details the use of anti-fungal proteins and traditional control methods.
- Highlights the role of deep learning tools for diagnosis and severity assessment.

Research and Data Resources

- Introduces curated biological databases, scientific literature, and reference collections.
- Provides access to SOPs, pathogen strains, and phenotypic data.
- Mentions tools for genomics, microbial profiling, and functional studies.

Technological Tools and Diagnostics

- Describes AI-based vision diagnostics using images from cameras, microscopes, and scanners.
- Explains models for binary classification and severity scoring.
- Supports rapid, multi-stage disease detection and environmental risk assessment.

Collaborative and Developmental Initiatives

- Encourages national and international research collaborations.
- Acknowledges scientific contributors and funding sources.
- Offers platform navigation, feedback options, and resource access pathways.